Analysis Report

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1. Methodology

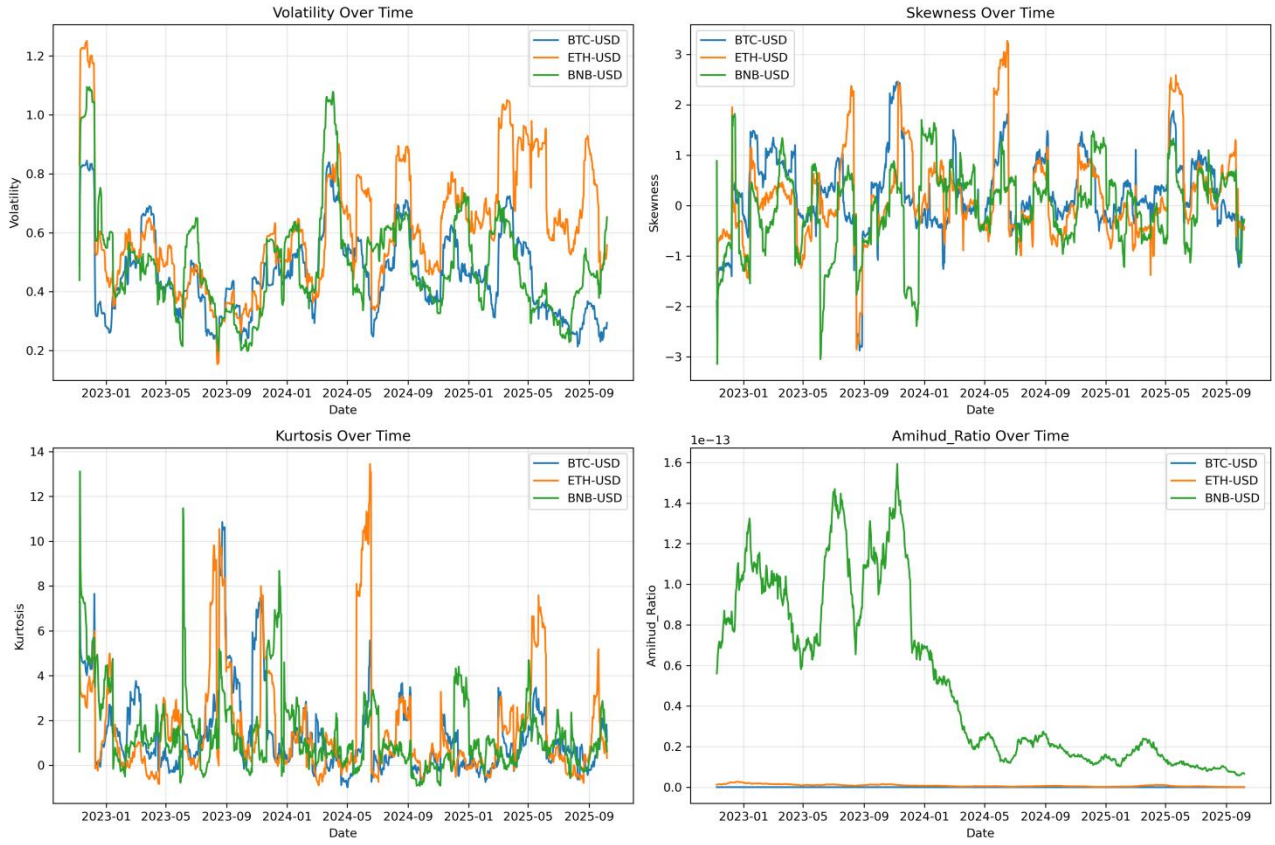
The analysis employs a risk factor analysis framework based on the code from the Jupyter Notebook file. Using a 30-day rolling window, we calculate four key risk metrics: annualized volatility, skewness, kurtosis, and the Amihud liquidity ratio. Logarithmic daily returns serve as the foundation for calculating volatility, skewness, and kurtosis, while the Amihud ratio measures price impact per unit of trading volume. We conduct correlation analysis to examine relationships between risk factors both within individual assets and across different cryptocurrencies. Multiple linear regression models are employed to test how these risk factors explain cryptocurrency returns, using standard Python data analysis libraries including pandas, numpy, scipy, and sklearn.

2. Data Sources

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and the Amihud liquidity ratio. Logarithmic daily returns serve as the foundation for calculating volatility, skewness, and kurtosis, while the Amihud ratio measures price impact per unit of trading volume. We conduct correlation analysis to examine relationships between risk factors both within individual assets and across different cryptocurrencies. Multiple linear regression models are employed to test how these risk factors explain cryptocurrency returns, using standard Python data analysis libraries including pandas, numpy, scipy, and sklearn.

Risk-related statistics line chart that changes over time is shown below:



3. Analysis Process

The analytical workflow begins with computing daily logarithmic returns from the raw price data. For each cryptocurrency, we calculate the four risk metrics using 30-day rolling windows to capture time-varying risk characteristics. The analysis proceeds on two levels: first, we examine correlations among different risk factors within each individual cryptocurrency through correlation matrices; second, we compare how the same risk factors move together across different cryptocurrencies. Finally, we implement multiple linear regression models where returns serve as the dependent variable and the four risk factors act as independent variables, allowing us to assess the statistical significance and explanatory power of each factor.

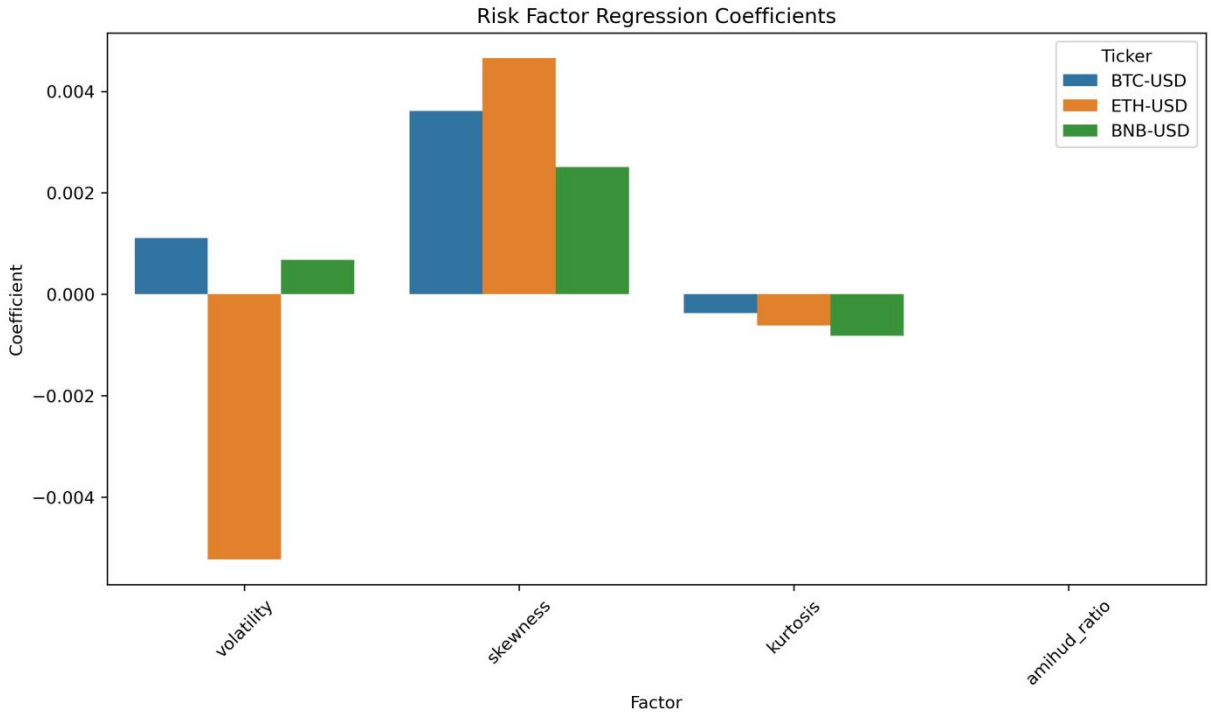
4. Risk Model Findings

The regression analysis reveals that skewness demonstrates statistically significant positive coefficients across all three cryptocurrencies. Specifically, BTC-USD shows a coefficient of 0.003617 ($p=0.0003$), ETH-USD 0.004657 ($p=0.0006$), and BNB-USD 0.002508 ($p=0.0290$). This pattern suggests that when return distributions exhibit right-skewness - indicating higher probability of extreme positive returns - investors appear willing to pay premiums, leading to subsequent positive returns. In contrast, volatility factors show no statistical

significance across all three assets, with BTC-USD at 0.001110 (p=0.8371), ETH-USD at -0.005232 (p=0.3109), and BNB-USD at 0.000675 (p=0.8973). Kurtosis factors similarly lack significance with coefficients of -0.000371 (p=0.3943), -0.000618 (p=0.2077), and -0.000825 (p=0.1464) respectively. The Amihud liquidity ratio coefficients approach zero and show complete statistical insignificance, indicating weak explanatory power for liquidity on daily returns.

Regarding model goodness-of-fit, all three regression models exhibit low R^2 values: 0.0131 for BTC-USD, 0.0121 for ETH-USD, and 0.0103 for BNB-USD. This indicates that the selected four risk factors explain only approximately 1% of return variations, with the majority of price movements driven by other factors. Correlation analysis reveals interesting patterns: volatility shows high correlation across assets, with coefficients generally exceeding 0.7, suggesting pronounced systematic risk characteristics in cryptocurrency markets. However, liquidity indicators demonstrate weaker cross-asset correlations, typically below 0.3, indicating that liquidity risk manifests more as asset-specific characteristics.

The coefficient bar chart is as follows：



5. Conclusions

Based on the empirical results, we draw several key conclusions. First, skewness emerges as the only risk factor demonstrating statistical significance across all three cryptocurrencies, supporting the skewness preference theory from behavioral finance, which suggests investors tend to chase assets with lottery-like return characteristics. Second, volatility - a core risk factor in traditional financial theory - fails to show significant pricing power in cryptocurrency markets, possibly reflecting unique risk preferences among investors in this emerging asset class. Third, the statistical insignificance of liquidity factors at daily frequencies may stem from two possibilities: either liquidity risk has been fully priced by the market and reflected in prices, or higher-frequency data is necessary to

capture its effects.

The low model goodness-of-fit (R^2 approximately 1%) indicates that relying solely on these four risk factors is insufficient to adequately explain cryptocurrency return variations. This suggests the importance of other influential factors such as market sentiment, network effects, regulatory developments, and technological advancements that may play more significant roles in cryptocurrency pricing. The cross-asset correlation patterns reveal that while volatility risk displays systematic characteristics, liquidity risk demonstrates stronger asset specificity, providing opportunities for risk diversification through portfolio construction. Notably, despite differences in market capitalization, technological foundations, and market positioning among the three cryptocurrencies, they exhibit similar risk-return relationship patterns, potentially reflecting common characteristics of cryptocurrency markets as an emerging asset class.

6. Risk Mitigation Recommendations

Based on our analytical findings, I recommend building our risk management approach around the specific patterns we observed in the data. Since skewness consistently emerged as the only statistically significant factor across all three cryptocurrencies, we should prioritize monitoring return distribution patterns and establish clear thresholds for

when skewness values indicate potential risk exposure. Given the low explanatory power of our current model—with R^2 values around just 1%—we need to recognize that these four traditional risk factors alone provide limited predictive capability, and therefore should be used as supplementary indicators rather than primary decision-making tools. The weak cross-asset correlations in liquidity measures suggest we should avoid making broad assumptions about liquidity conditions across different cryptocurrencies, instead developing individualized liquidity assessment protocols for each major asset we track.